

MAURO FACCIN

Curriculum Vitæ

Dip. Fisica e Astronomia
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Preferred pronouns: he/him

ACTUAL POSITION

2024-today **DFA at University of Padova** University of Padova, Italy
Interests/Research line

- Analysis of climate data and their interaction with human made networks
- Mesoscopic structures and their influence on the dynamical system.

PROFESSIONAL EXPERIENCE

2023 - 2024 **University of Bologna, (DIMEC and INFN)** Bologna, Italy
Postdoctoral researcher

- Analysis of medical data through the use of Machine learning and complex networks combined approaches.
- Prediction of disease survival
- Analysis of information dynamics in social networks and medical systems.

2022 - 2023 **Istituto Professionale Statale "G. B. Garbin"** Schio, Italy
High school math teacher

2021 - 2022 **Université de Paris, (CEPED and IRD)** Paris, France
Postdoctoral researcher

- Vaccines hesitancy unveiled through hypergraph analysis.
- We analyse the discussion on microblogging services as a dynamical system spreading on a hypergraph.
- Analysis of the textual content and its dynamical and geographic properties.

2020 - 2021 **University of Turin (Computer Science)** Torino (Italy)
Visiting

2015 - 2020 **Université Catholique de Louvain (ICTEAM)** Louvain la Neuve, Belgium
MOVE IN Louvain Research Fellow

- Interplay between complex networks and dynamical systems.
- Making use of mathematical tools from information theory we could analyse dynamical systems, their evolution and the emerging mesoscopic structures.
- We successfully built a bridge between statistical inference and dynamical approaches in community or block detection.
- Development of a `python` library.
- Spatial analysis of disease spreading with compartmental models. (tuberculosis incidence inference in African countries)

2011 - 2015 **ISI Foundation** Torino, Italy
Junior researcher

- Quantum meets complex networks.
- I contributed to a multidisciplinary team interested in using tools from complex networks to study quantum system.

2006 - 2011 **Bifi Institute** Zaragoza, Spain
PhD

- Use of Ising-like models to characterize biopolymers with application to protein folding and protein sequencing.
- Development and release of corresponding software

FELLOWSHIPS AND PROJECTS

2021 - 2022 **TRACTRUST (ANR-20-COVI-0102)**
2017 - 2021 **Innoviris Iris-TB (D1.31402.007-F)**

2015 - 2017 MOVE-IN Louvain Fellowship (with support of Marie Curie Actions)

PROGRAM COMMITTEES

1 Dec 2021	Complex Networks 2021	Madrid, Spain
1 Dec 2020	Complex Networks 2020	Madrid, Spain
10 Dec 2019	Complex Networks 2019	Lisbon, Portugal
22 Feb 2019	BeNet 2019	Hasselt, Belgium
11 Dec 2018	Complex Networks 2018	Cambridge, UK
29 Nov 2017	Complex Networks 2017	Lyon, France
30 Nov 2016	Complex Networks 2016	Milan, Italy

ORGANIZING COMMITTEES

25 Oct 2024	Networks Days Local committee	Padova, Italy
August 2019	Network Reconstruction Science-athon Coorganized with Alexandre Bovet, Benjamin Chiem, Leo Gutierrez, Alexey Medvedev, Leto Peel, Mridul Seth	Beaumont, Belgium
15 Dec 2016	Belgian Network Research Meeting (BeNet) 2016 Coorganized with Sophie Béreau, Jean-Charles Delvenne, Yerali Gandica, Leo Gutierrez, Leto Peel, Michael Schaub	Louvain-la-Neuve, Belgium
21 Sep 2016	Conference on Complex Systems 2016 workshop on Coarse-graining Complex Systems Coorganized with Leto Peel, Renaud Lambiotte, Michael Schaub	CCS2016, Amsterdam, Netherlands
2 Jun 2014	Quantum Frontiers for Network Science Coorganized with Filippo Radicchi and Zoltan Zimborás	NetSci 2014, Berkeley, California

REVIEWER

- **Advances in Complex Systems**
World Scientific
- **Applied Network Science**
Springer Nature
- **Entropy**
Multidisciplinary Digital Publishing Institute (MDPI)
- **Journal of Complex Networks**
Oxford University Press
- **Journal of Statistical Mechanics**
IOP
- **Natural Computing**
Springer Netherlands
- **PLOS ONE**
Public Library of Science (PLOS)
- **PRA, PRB, PRE, PRL, PRX**
American Physical Society
- **Science Advances**
American Association for the Advancement of Science
- **Scientific Reports**
Nature Portfolio

TEACHING

2023-2024	DAT301 Social Network Analysis with F. Gargiulo and A. Apolloni	University Mohammed VI Polytechnic
2015-2020	LINMA2111 Algorithms and Complexity with J-C Delvenne	Université Catholique de Louvain

- 2005 **Physics and Math at primary school** Hetauda, Nepal
NEST
- 2004 **guide in Science Museum** Padova, Italy
Organizer: INFN

STUDENT ADVISING

- 2024 **N. Barbieri** University of Bologna
Corelatore tesi magistrale with F. Durazzi and D. Remondini in Applied Physics
Integration of artificial intelligence and network approaches to classify data in social networks
- 2019 **G. Izzì** University of Turin
Tesi magistrale with A. Semeraro and G. Ruffo (informal tutor). Computer Science
Analysis of temporal patterns and forecasting of popularity for programming languages on Stack Overflow platform
- 2016 **H. Devillez** UCLouvain
Applied Maths (project with L. Peel)
- 2016 **M. Huet** UCLouvain
Elec. Eng. (project with L. Peel)

EDUCATION

- 2020 **24CFU** Univesità telematica Pegaso
Discipline antropo-psico-pedagogiche e in metodologie e tecnologie didattiche
- 14 Sept 2012 **PhD in Physics** University of Zaragoza
supervisor: prof. Pierpaolo Bruscolini
Analysis and Development of Algorithms for the Characterization of Biopolymers and Application to Protein Sequencing and Protein Folding
- 15 Oct 2008 **Master in Physics and Technology** University of Zaragoza
supervisor: prof. Pierpaolo Bruscolini
Coevolution of Strategy and Population Structure in the Prisoner Dilemma
- 10 Oct 2005 **MSc in Physics** University of Padova
supervisor: prof. Enzo Orlandini
Beris Edwards Model Simulations for Liquid Crystals in Oscillatory Poiseuille Flux

MAINTAINED PROJECTS

- **AISA**
Python module aimed at partitioning a network through Autoinformation maximization
- **CartoTB**
Cartography of predicted tuberculosis in high burden settings

PREPRINTS

- **Temporal and geographic analysis of the Hydroxychloroquine controversy in the French Twittosphere** arXiv:2304.14075
M. Faccin, E. Schultz and F. Gargiulo

PUBLICATIONS

- **Analyzing Satellite Imagery to Target Tuberculosis Control Interventions in Densely Urbanized Areas of Kigali, Rwanda: Cross-Sectional Pilot Study**
JMIR Public Health Surveill 2025;11:e68355
M. Faccin, C. Geenen, M. Happaerts, S. Ombelet, P. Migambi, E. André
- **Mapping of regions with low tuberculosis notification and estimation of diagnostic gaps in Cameroon evidence from OpenStreetMap and WorldPop data**
Scientific Reports, 15, Article number: 4992 (2025)
C.G. Jouego, T.N. Collins, B. Tegomoh, C. Geenen, P.M. Netongo, E. André, M. Faccin

- **Measuring dynamical systems on directed hypergraphs** Phys Rev E, 106 (034306), 2022
M. Faccin
- **Assessing the influence of French vaccine critics during the two first years of the COVID-19 pandemic** PLOS ONE, 17(8): e0271157
M. Faccin, F. Gargiulo, L. Atlani-Duault, J.K. Ward
- **Data-driven identification of communities with high levels of tuberculosis infection in the Democratic Republic of Congo.** Sci Rep 12, 3912 (2022)
M. Faccin, O. Rusumba, A. Ushindi, M. Riziki, T. Habiragi, F. Boutachkourt, E. André
- **State aggregations in Markov chains and block models of networks** Phys. Rev. Lett., 127, 078301 (2021)
M. Faccin, M.T. Schaub and J.-C. Delvenne
Figure selected for the journal cover
- **Complex Networks from Classical to Quantum** Communications Physics 2-53 (2019)
J. Biamonte, M. Faccin, M. De Domenico
- **Outbreak of multidrug-resistant tuberculosis in South Africa undetected by WHO-endorsed commercial tests: an observational study** The Lancet Infectious Diseases 2018
Makhado, Matabane, Faccin, Pinçon, Jouet, Boutachkourt, Goeminne, Gaudin, Maphalala, Beckert, Niemann, Delvenne, Delmée, Razwiedani, Nchabeleng, Supply, de Jong, André
- **Entrograms and coarse graining of dynamics on complex networks** Journal of Complex Networks
M. Faccin, M.T. Schaub, J.-C. Delvenne
- **Chiral Quantum Walks** Phys. Rev. A 93, 042302
D. L. Lu, J. D. Biamonte, J. Li, H. Li, T. H. Johnson, V. Bergholm, M. Faccin, Z. Zimborás, R. Laflamme, J. Baugh, and S. Lloyd
- **Mapping the topography of a protein energy landscape** J. Am. Chem. Soc., 2015, 137 (46), pp 14610–14625
R. D. Hutton, J. Wilkinson, M. Faccin, E. M. Sivertsson, A. Pelizzola, A. R. Lowe, P. Bruscolini, L. S. Itzhaki
- **Community Detection in Quantum Complex Networks** Phys. Rev. X 4 (4 2014), p. 041012
M. Faccin, P. Migdal, T. H. Johnson, V. Bergholm, and J. D. Biamonte
- **Degree Distribution in Quantum Walks on Complex Networks** Phys. Rev. X 3 (4 2013), p. 041007
M. Faccin, T. Johnson, J. Biamonte, S. Kais, and P. Migdal
- **Quantum Transport Enhancement by Time-Reversal Symmetry Breaking** Sci. Rep. 3 (2013). Ed. by NPG, p. 2361
Z. Zimboras, M. Faccin, Z. Kadar, J. Whitfield, B. Lanyon, and J. Biamonte
- **MS/MS Spectra Interpretation as a Statistical-Mechanics Problem** Analytical Chemistry 85.10 (2013), pp. 4884–4892
M. Faccin and P. Bruscolini
- **Ground-state spin logic** Europhysics Letters 99.5 (2012), p. 57004
J. D. Whitfield, M. Faccin, and J. D. Biamonte
- **Analysis of the equilibrium and kinetics of the ankyrin repeat protein myotrophin** The Journal of Chemical Physics 134.7, 075102 (2011), p. 075102
M. Faccin, P. Bruscolini, and A. Pelizzola

CONFERENCES (ORAL CONTRIBUTIONS)

- 22 Jul 2025 **Climate-induced delay propagation in rail networks as a non-Markovian process**
StatPhys29 (Complex Networks)
M. Faccin, T. Scagliarini, M. De Domenico
- 10 Oct 2023 **International topic distribution in the HCQ controversy on Twitter**
CCS/Italy @ Napoli (Italy)
M. Faccin, E. Schultz, F. Gargiulo
- 7 Mar 2023 **Dynamical systems as a mean for complex systems analysis**
seminar at the University of Bologna
M. Faccin

- 9 Nov 2022 **Data-driven mapping of tuberculosis incidence allows identifying communities with high circulation of tuberculosis in the Democratic Republic of Congo**
The Union Conference 2022 – Online
M. Faccin, C. Nsop Mulaj, E. Swana Kawana, T. Yav Ikum, E. André
- 9 Nov 2022 **Identification of urban areas presenting high incidence of tuberculosis using satellite images allows increasing the yield of active case finding in Kigali, Rwanda.**
The Union Conference 2022 – Online
M. Faccin, P. Uwizeye, F. Nsabimana, P. Migambi, E. André
- 17 Oct 2022 **Use of compartmental models to detect high TB incidence rates in South-Kivu (DRC)**
CCS 2022 @ Palma de Mallorca/Online
M. Faccin, E. André
- 20 June 2022 **Reach of vaccine-critical content on Twitter during the COVID-19 pandemic**
French Regional Conf. on Complex Systems (FRCCS) @ Paris
M. Faccin, F. Gargiulo
Given by F. Gargiulo
- 30 May 2022 **Outreach of vaccine-critic content on French-speaking Twitter**
CompleNet 2022 – Online
M. Faccin, F. Gargiulo, J. Ward
- 2 Dec 2021 **Autoinformation state aggregation**
Complex Networks 2021 @ Madrid (Spain)
M. Faccin, M. T. Schaub, J-C. Delvenne
- 30 Nov 2021 **Vaccine-critical discourse on French-speaking Twitter and its outreach**
Complex Networks 2021 @ Madrid (Spain)
M. Faccin, F. Gargiulo, J. Ward
Best lightning talk award
- 19 Oct 2021 **State aggregation for dynamical systems: An information-theoretic approach**
Math Dept, Oxford University, (UK – Online)
Network Seminar 2021
- 7-9 Dec 2020 **Using predictive models and risk assessment tools to find unreported TB cases**
Washington DC, USA (online)
Global Digital Health Forum 2020
- 20-24 Oct 2020 **Data-driven TB active-case finding in South Kivu (DRC)**
Paris, France (online)
The Union 2020
- 10-12 Dec 2019 **Auto-information in non-Markovian diffusion systems**
Lisbon, Portugal
Complex Networks 2019
- 22 Feb. 2019 **Variable aggregation for dynamical systems on networks**
Hasselt, Belgium
Benet 2019
- 23-28 Sept. 2018 **A dynamical approach to block structure analysis**
Thessaloniki, Greece
CCS 2018
- 11-15 Jun 2018 **Dynamical Markov Modules**
Paris, France
Netsci 2018
- 21-24 Mar 2017 **Community detection in networks with non-Markovian dynamics**
Dubrovnik, Croatia
CompleNet 2017
- 30 Nov-2 Dec 2016 **Behind Communities: Dynamics Define Modules**
Milan, Italy
ComplexNetworks 2016
- 19-22 Sep 2016 **Beyond Communities: Dynamical Markov Modules**
Amsterdam, Netherlands
CCS 2016
- May 30-June 3 2016 **Mesoscopic Structure And Diffusion Process Memory**
Seoul, Korea
NetSci 2016
- 10-15 Jan 2015 **Mesoscopic Structures and Diffusion Process Memory**
Wrocław, Poland
NetSci-X 2016
- 2-6 Jun 2015 **Structural controllability of EEG temporal causal networks**
Zaragoza, Spain
NetSci 2015
- 22-26 Sep 2014 **Quantum Approaches to Community Detection**
Lucca, Italy
ECCS 2014
- 12-14 Mar 2014 **Community detection on quantum systems**
Bologna, Italy
CompleNet 2014
- 22-24 Jan 2014 **Community Detection on Quantum Systems**
Zaragoza, Spain
BIFI2014
- 24-26 May 2013 **Degree distribution in quantum complex networks**
Waterloo, Canada
IQC workshop on quantum computation and complex networks
- 6-8 Nov 2012 **Ground state spin logic**
Torino, Italy
Tensor Network States and Algebraic Geometry Workshop

- 2-4 Feb 2011 **Equilibrium and kinetics of the ankyrin repeat protein Myotrophin through a simple statistical mechanical model** Zaragoza, Spain
V BIFI National Conference
- 5-7 Feb 2009 **Bayesian Characterization of Protein Tandem Mass Spectra** Zaragoza, Spain
IV BIFI National Conference
- 8-9 May 2008 **Noise description in MSMS mass spectrometry, looking for a de novo sequencing algorithm** San Sebastian, Spain
VI Jornadas Aragonesas de Estabilidad y Plegamiento de Proteínas

CONFERENCES (POSTER CONTRIBUTIONS)

- 1 Dec 2021 **Directed hyper-graphs and dynamical systems** Complex Networks 2021 @ Madrid (Spain)
M. Faccin, F. Gargiulo
- 17-25 Sept 2020 **State aggregation in non-Markovian systems by auto-information maximization** Rome, Italy
Netsci 2020
- 17-25 Sept 2020 **The variability of network structures inferred from time series data** Rome, Italy
Netsci 2020
- 10-12 Dec 2019 **The variability of network structures inferred from time series data** Lisbon, Portugal
Complex Networks 2019
- 1-3 July 2019 **Variable aggregation on (broadcasting) networks** Trento, Italy
CCS/Italy 2019
- 23-25 Mar 2016 **Memory and Mesoscopic Structures in Diffusion Processes** Dijon, France
CompleNet 2016
- 7-10 Jul 2010 **Folding Pathways Characterization for Ankyrin Repeat Protein Myotrophin** Zaragoza, Spain
IV Spanish Portuguese Biophysical Congress
- 2-6 Jun 2008 **Statistical characterization of fragment ions in MSMS mass spectrometry** Sitges, Spain
XXI Sitges Conference on Statistical Mechanics: Statistical Mechanics of Molecular Biophysics

COMPUTER SKILLS

- Coding **Python, C, Matlab, Fortran**
- Shell **Bash, iPython/Jupyter**
- Versioning **GIT, Subversion, Mercurial**
- Plotting **Matplotlib (python), Gnuplot, PGFPlots (LaTeX)**
- Typesetting **LaTeX, LibreOffice, MSOffice, Markdown**
- Imaging **Gimp (Raster), Tikz (LaTeX), Inkscape (Vectorial), Rawtherapee (RAW)**
- Web **HTML5, CSS/SCSS, static site generators, Wordpress**

LANGUAGES

- Italian **Mother tongue**
- English **Fluent**
- Spanish **Fluent**
- French **Fair**

Mauro Faccin